

Daniel Navarro

Data Scientist & AI Engineer | RAG/LLM Systems, Data Engineering & BI | MA Public Policy (CEU)

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danielnavarromadrinan.com | Geneva, Switzerland (open to relocation)

I build data and AI systems end to end, from scraping and ETL to **RAG pipelines**, models and the dashboards people decide with. I combine applied **machine learning** (tree-based, time series, clustering) and modern **AI** (RAG, vector and graph databases, local LLMs) with production-minded pipelines in **Python, SQL, Docker and AWS/Azure**, for private-sector clients and for the UN in Geneva. My training in anthropology and quantitative policy analysis means I build for the people who use the output, and explain it clearly to technical and non-technical audiences.

CORE COMPETENCIES

Applied AI & RAG: RAG pipelines, vector and graph databases, embeddings, knowledge graphs, semantic search, LLMs (Hugging Face, local models via Docker Model Runner), NLP, automated report generation; agentic coding with **Claude Code**.

Machine Learning & Statistics: Python (pandas, scikit-learn), R; supervised and unsupervised ML (tree-based, regression, clustering), time series (ARIMA, XGBoost, NARX), deep learning, topic models and transformer sentiment.

Data Engineering & Cloud: ETL pipelines, web scraping, REST APIs, n8n automation, data architecture (ERDs, schemas, user flows); SQL (PostgreSQL, SQLite), Docker, AWS, Azure, Git.

BI & Visualisation: Power BI, Preset, Dash, Plotly, custom CRMs, KPI reporting and data storytelling; QGIS, GeoJSON, Copernicus/Sentinel satellite data.

Quantitative Policy & Research: impact evaluation and causal inference (TWFE, event studies), survey-weighted regression, procurement and corruption-risk analytics, MEAL.

PROFESSIONAL EXPERIENCE

Instrumental Partners LLC

Data Engineer & BI Consultant

Geneva (remote)

Jul 2025–Present

- Built an **AI RAG and graph knowledge base** with automated competitor and market-intelligence reporting over **1M+ federal procurement contracts** and 8k+ competitors, using proprietary RAG pipelines and local LLMs.
- Engineered Python web-scraping, API and **ETL pipelines** structuring procurement data (FPDS, DIBBS), and designed the data architecture from ERDs and schemas to user flows.
- Applied ML and statistical models for **pricing and bidding recommendations** in a custom CRM, with Apollo API enrichment for prospecting workflows.
- Built the **BI dashboards** clients run on (Preset, Power BI, SQLite), and shipped the firm's website end to end (content, case studies, SEO) with n8n content and news-monitoring automation.

UN OCHA

Information Management Intern

Geneva, Switzerland

Jun 2026–Present

- Built an **automated AI report-generation system** (not a chatbot): it monitors news, social and API sources and builds a fact and relationship graph of events, people and organisations (NLP, graph and vector databases, embeddings), so every statement is traceable to a named source.
- Containerised the UI with **Docker** for deployment on **Azure**; trained country offices to use and extend the system and wrote the accompanying guides.
- Contributed to Flash Appeal and Situation Analysis products and needs analysis for the **Venezuela 2026 earthquake**, mapping populated areas to surface overlooked populations under daily deadlines.

World Bank / CEU

Researcher & Workstream Lead – Educational Inequality (TIMSS)

Vienna, Austria

Sep 2025–May 2026

- Led the CEU team on a multi-country World Bank project (task allocation and milestones in Microsoft Planner) and was the interface with World Bank counterparts on deliverables.
- Cleaned and merged large-scale TIMSS data; ran **survey-weighted regression** and comparative regional analysis of socio-economic gaps with a reproducible methodology.

Independent <i>Independent Consultant – Data Science & GIS</i>	Bogotá, Colombia <i>Jul 2024–Jul 2025</i>
- Built ETL pipelines and time-series models (ARIMA, XGBoost, NARX) for air-quality and climate forecasting on Copernicus/Sentinel data; deployed dashboards in Dash, Plotly and Power BI. - Created GeoJSON datasets for FM-radio coverage, environmental monitoring and infrastructure planning.	
EU Delegation in Colombia <i>Administrative Assistant / Archivist</i>	Bogotá, Colombia <i>May 2024–Oct 2024</i>
Designed and executed a five-month plan over 20 years of EU cooperation records (about 8 tons, 9,300 files); built Excel tracking that forecast progress with ~90% accuracy and met 100% of the target.	
Organization of American States (OAS) <i>National Office Intern (Research) & International Electoral Observer</i>	Santo Domingo & El Salvador <i>Jun 2023–Feb 2024</i>
- Analysed migration discourse by scraping presidential speeches and social-media data with Python NLP; drafted analytical reports and ran a digital strategy that lifted engagement by 500%+. - Observed the 2024 Salvadoran elections (DECO mission), documenting practices with a legal and gender focus.	
I2T Laboratory & SegLab, Universidad ICESI <i>Research Assistant / Data Analyst</i>	Cali, Colombia <i>Feb 2021–Jul 2024</i>
- Applied ML and deep learning to geospatial health data linking environmental exposure to Parkinson's disease progression, a 25% improvement over baseline; presented to doctors, engineers and designers. - Security and peacebuilding research with QGIS and Power BI dashboards feeding policy papers and articles.	

EDUCATION

Central European University (CEU) <i>MA in Public Policy – Quantitative Policy Analysis</i>	Vienna, Austria <i>Sep 2025–Jun 2026</i>
Thesis: <i>(I'm)Paz Total: Quantitative Evaluation of the Política de Paz Total in Colombia</i> . Coursework: Text Analysis & NLP , Impact Evaluation (R), Public Procurement Analytics, Data Visualization.	
Universidad ICESI <i>BA in Anthropology (Antropólogo), Magna Cum Laude</i>	Cali, Colombia <i>Feb 2019–Feb 2024</i>
GPA 4.5/5.0, ranked 1st in class, Dean's List. Erasmus exchange, University of Turin (2022).	

SELECTED PROJECTS (GITHUB.COM/DANIEL-NAVARRO-M)

- Paz Total impact evaluation:** monthly panel from National Police and ACLED data for three conflict regions; two-way fixed effects and event studies in R.
- Rail Baltica corruption-risk analytics:** contract-level Corruption Risk Index from red-flag indicators validated by logistic regression on EU tender data (with P. Biša).
- Migration discourse NLP:** seeded LDA, DBSCAN and transformer sentiment on 43 presidential speeches and ~8,800 Reddit sentences.

PUBLICATIONS, AWARDS & LANGUAGES

Publication: Milanese & Navarro (2025). *Beyond municipal boundaries: metropolitanization of electoral behavior in Cali using k-means clustering*. Revista Geographia (Brazil).

Awards & Certifications: **1st Place**, 2nd TechPolítica hackathon, FLACSO Ecuador • MEAL (Monitoring, Evaluation, Accountability & Learning) • Satellite Image Processing • Effective Project Planning & Evaluation in Health Research.

Languages: Spanish (native) • English (C1, IELTS Academic 8.0) • Italian (B1) • French (A2) • German (beginner).